

Lecture-17

Mathematics 1 (15B11MA111)

CO [C105.4]

Module: Vector Differentiation
Topic: Curves, limit, Continuity and differentiability

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Vector Differential and Integral Calculus

Scalar function A scalar function $f(x, y, z)$ is a function defined at each point in a certain domain D in space. Its value is real and depends only on the point $P(x, y, z)$ in a space, but not on any particular coordinate system being used. For example, The distance function in 3-D space which defines the distance between the points $P(x, y, z)$ and $P_0(x_0, y_0, z_0)$

$$f(P) = f(x, y, z) = \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}$$

defines a scalar field.

Vector function A function $\bar{v} = \bar{v}(P) = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$ defined at each point $P \in D$ is called a vector function. In Cartesian coordinates, we can write

$$\bar{v} = v_1(x, y, z)\hat{i} + v_2(x, y, z)\hat{j} + v_3(x, y, z)\hat{k}$$

For example, the velocity field $\bar{v}(P)$ defined at any point P on a rotating body defines a vector field.

Level surface Let $f(x, y, z)$ be a single valued continuous scalar function defined at every point $P \in D$. Then $f(x, y, z) = c$, a constant, defines the equation of a surface and is called the level surface of the function.

Example 1. Find the level surface of the scalar fields in space, defined by the following functions

$$(i) f(x, y, z) = (x^2 + y^2 + z^2), \quad (ii) f(x, y, z) = z - \sqrt{x^2 + y^2}$$

Solution

(i) We find that $f(x, y, z) = c$ gives $x^2 + y^2 + z^2 = c$.

Therefore the level surfaces are spheres of radius $\sqrt{c}$.

(ii) We find that $f(x, y, z) = c$ gives $z - \sqrt{x^2 + y^2} = c$ or $x^2 + y^2 = (z - c)^2$. The level surfaces are cones.

Parametric representation of curves

A curve C in the two dimensional xy -plane can be parameterized by $x = x(t), y = y(t), a \leq t \leq b$. Then, the position vector of a point P on the curve C can be written as

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j}.$$

Similarly, a three-dimensional curve or a space curve C can be parameterized as

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}, a \leq t \leq b.$$

Circle The parametric form of the circle $x^2 + y^2 = a^2$, can be written as $\vec{r}(t) = a \cos t \hat{i} + a \sin t \hat{j}, 0 \leq t \leq 2\pi$.

Ellipse The parametric form of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ can be written as $\vec{r}(t) = a \cos t \hat{i} + b \sin t \hat{j}, 0 \leq t \leq 2\pi$.

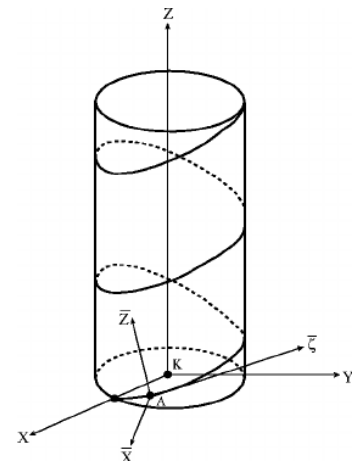
Circle in a plane in 3-dimensions Consider the circle as $x^2 + y^2 = a^2, z = d$ which lies in the plane $z = d$. Then, we can write the parametric form as

$$\vec{r}(t) = a \cos t \hat{i} + a \sin t \hat{j} + d \hat{k}.$$

Helix The curve traced by the vector function

$$\vec{r}(t) = a \cos t \hat{i} + a \sin t \hat{j} + ct \hat{k}, a > 0$$

is called a circular helix.



Line in space In a 2D-plane, a line is determined by a point and slope of line as $y - y_0 = m(x - x_0)$.

In 3D-space, a line is determined by a point and a vector giving the direction of line.

Let L be a line in a space passing through a point $P_0(x_0, y_0, z_0)$ parallel to a vector $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$; then L is the set of all points $P(x, y, z)$ for which $\overrightarrow{P_0P}$ is parallel is $\vec{v}$.

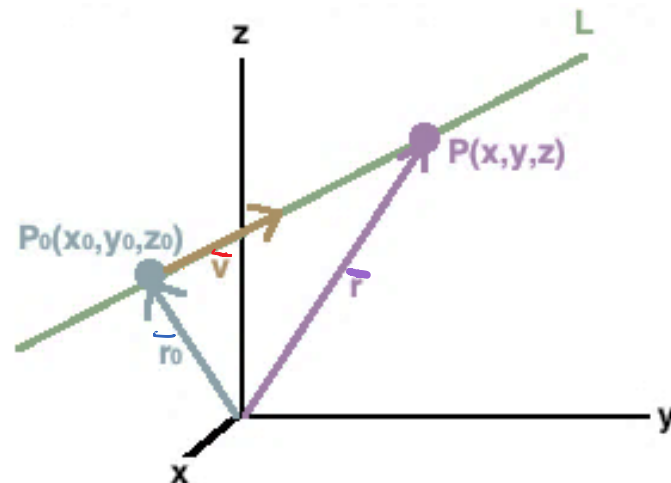


Image from: <http://mathonline.wikidot.com/>

$$\therefore \overrightarrow{P_0P} = t \bar{v} \quad [\text{for some scalar } t, \text{value of } t \text{ depends on the location of } P \text{ on line.}]$$

$$\Rightarrow (x - x_0)\hat{i} + (y - y_0)\hat{j} + (z - z_0)\hat{k} = t(v_1\hat{i} + v_2\hat{j} + v_3\hat{k})$$

$$\Rightarrow x\hat{i} + y\hat{j} + z\hat{k} = x_0\hat{i} + y_0\hat{j} + z_0\hat{k} + t(v_1\hat{i} + v_2\hat{j} + v_3\hat{k})$$

$$\Rightarrow \bar{r}(t) = \bar{r}(t_0) + t \bar{v}$$

$$\text{or } \bar{r}(t) = \bar{r}_0 + t \bar{v}; -\infty < t < \infty$$

This is vector equation of line through P_0 and parallel to $\bar{v}$.

Parametric equation for above line

$$x = x_0 + tv_1, y = y_0 + tv_2, z = z_0 + tv_3$$

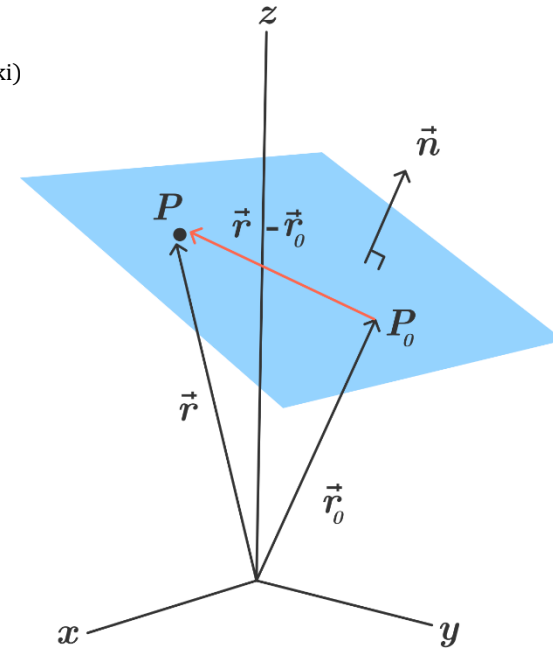
Example 2. Find parametric equation for a line through $(-2, 0, 4)$ parallel to $\bar{v} = 2\hat{i} + 4\hat{j} - 2\hat{k}$.

Solution $x = -2 + 2t, y = 4t, z = 4 - 2t$.

Equation of a plane in space

A plane in space is determined by a point on the plane and its “tilt” or orientation. This tilt is defined by a vector **normal to plane**.

(Image from: <https://brilliant.org/wiki>)



Plane is set of points for which $\overrightarrow{P_0P}$ is orthogonal to $\vec{n}$.

$$\Rightarrow \vec{n} \cdot \overrightarrow{P_0P} = 0 \text{ or } (\vec{r} - \vec{r}_0) \cdot \vec{n} = 0$$

$$\Rightarrow (A\hat{i} + B\hat{j} + C\hat{k}) \cdot ((x - x_0)\hat{i} + (y - y_0)\hat{j} + (z - z_0)\hat{k}) = 0$$

$$\Rightarrow A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Example 3. Plane through $(-3, 0, 7)$ perpendicular to $5\hat{i} + 2\hat{j} - \hat{k}$.

Solution $5[x - (-3)] + 2(y - 0) + (-1)(z - 7) = 0$

$$5x + 2y - z = -22.$$

Note.

1. Vector $\bar{n} = A\hat{i} + B\hat{j} + C\hat{k}$ is normal to plane $Ax + By + Cz = D$.
2. For a plane through three points A, B and C,

$$\bar{n} = \overrightarrow{AB} \times \overrightarrow{AC}$$

Limit, continuity and differentiability of vector functions

The concepts of limit, continuity and differentiability of calculus can easily be used for vector functions. Let the vector function be written in its parametric form, that is $\vec{v} = \vec{v}(t)$. In the Cartesian system, we can write $\vec{v}(t) = v_1(t)\hat{i} + v_2(t)\hat{j} + v_3(t)\hat{k}$. Then, we define the following.

Limit The vector function $\vec{v}(t)$ has the limit $\vec{l}$ as $t \rightarrow a$, if $\vec{v}(t)$ is defined in some neighborhood of a , except possibly at $t = a$ and $\lim_{t \rightarrow a} |\vec{v}(t) - \vec{l}| = 0$. We write $\lim_{t \rightarrow a} \vec{v}(t) = \vec{l}$.

In the Cartesian system, this implies that limits of the component functions $v_1(t)$, $v_2(t)$ and $v_3(t)$ exist as $t \rightarrow a$ and

$$\lim_{t \rightarrow a} v_1(t) = l_1, \lim_{t \rightarrow a} v_2(t) = l_2, \lim_{t \rightarrow a} v_3(t) = l_3$$

where $\bar{l} = l_1 \hat{i} + l_2 \hat{j} + l_3 \hat{k}$.

Continuity A vector function $\bar{v}(t)$ is said to be continuous at $t = a$, if (i) $\bar{v}(t)$ is defined in some neighborhood of a ,

(ii) $\lim_{t \rightarrow a} \bar{v}(t) = \bar{v}(a)$.

In Cartesian system, this implies that $\bar{v}(t)$ is continuous at $t = a$, if and only if the component functions $v_1(t)$, $v_2(t)$ and $v_3(t)$ are continuous at $t = a$.

Differentiability A vector function $\bar{v}(t)$ is said to be differentiable at a point, if the limit

$$\lim_{\Delta t \rightarrow 0} \frac{\bar{v}(t + \Delta t) - \bar{v}(t)}{\Delta t}$$

exists. If the limit exists, then we write it as $\bar{v}'(t)$ or as $d\bar{v}/dt$. In the Cartesian system, this implies that the component functions $v_1(t)$, $v_2(t)$ and $v_3(t)$ are differentiable at a point t , that is the limits

$$\lim_{\Delta t \rightarrow 0} \frac{v_i(t + \Delta t) - v_i(t)}{\Delta t}, i = 1, 2, 3 \text{ exist.}$$

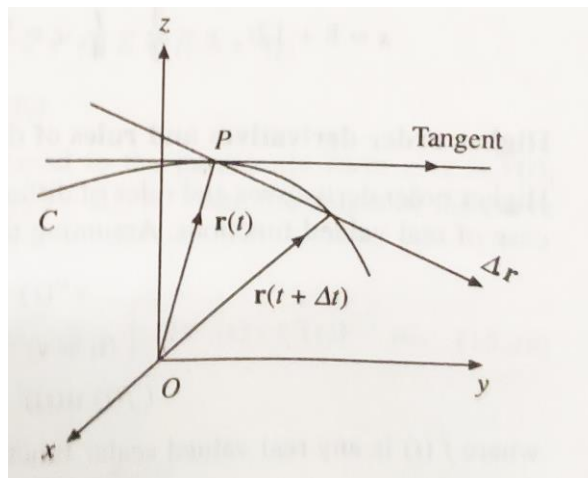
Therefore, $\bar{v}'(t) = v'_1(t)\hat{i} + v'_2(t)\hat{j} + v'_3(t)\hat{k}$.

Let $\bar{v}(t) = \bar{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ be the parametric representation of a curve C. Then

$$\frac{d\bar{r}}{dt} = \bar{r}'(t) = \frac{dx(t)}{dt}\hat{i} + \frac{dy(t)}{dt}\hat{j} + \frac{dz(t)}{dt}\hat{k}.$$

Geometric representation of $\frac{d\bar{r}}{dt}$: $\frac{d\bar{r}}{dt}$ is the tangent line to the curve at

the point P. So, the unit tangent vector is given by $\frac{d\bar{r}}{dt} / \left| \frac{d\bar{r}}{dt} \right|$.



Smooth curve Let $\vec{r}(t)$ denotes the position vector of a point P on the curve C. Let $\vec{r}(t)$ have continuous first derivative, that is the component function $x(t), y(t)$ and $z(t)$ have continuous first derivatives. Let $\vec{r}'(t) \neq 0$, for all $t \in (a, b)$. Then, $\vec{r}(t)$ defines a smooth function on (a, b) . The curve C traced by $\vec{r}(t)$ is called a smooth curve.

Example 4. Represent the parabola $y = 1 - 2x^2, -1 \leq x \leq 1$ in parametric form. Hence, find $\vec{r}'(0)$ and $\vec{r}'(\frac{1}{\sqrt{2}})$.

Solution Let $x = \sin t$. then $y = 1 - 2\sin^2 t = \cos 2t$, and $-\pi/2 \leq t \leq \pi/2$. Hence,

$$\bar{r}(t) = \sin t \hat{i} + \cos 2t \hat{j}, -\pi/2 \leq t \leq \pi/2.$$

For $x = 1/\sqrt{2}$, we get $t = \pi/4$. Therefore,

$$\bar{r}'(t) = \cos t \hat{i} - 2 \sin 2t \hat{j} \quad \text{and} \quad \bar{r}'(0) = \hat{i}, \bar{r}'(\pi/4) = \left(\frac{\hat{i}}{\sqrt{2}}\right) - 2\hat{j}.$$

The tangent line at $t = 0$ is parallel to the x - *axis*. (Note that $t = 0$ gives $x = 0, y = 1$ which is the vertex of the parabola.)

Example 5. Find the tangent vector to the curve whose parametric representation is

$x = \cos t, y = \sin t, z = t, \pi \leq t \leq \pi$. Hence, find the unit tangent vector.

Solution the position vector of a point on the curve is

$\vec{r}(t) = \cos t \hat{i} + \sin t \hat{j} + t \hat{k}$. Therefore, the tangent vector is

$$\vec{r}'(t) = -\sin t \hat{i} + \cos t \hat{j} + \hat{k}.$$

We have $|\vec{r}'(t)| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$.

Hence, the unit tangent vector is

$$\hat{r}'(t) = (-\sin t \hat{i} + \cos t \hat{j} + \hat{k})/\sqrt{2}.$$

THANK YOU